

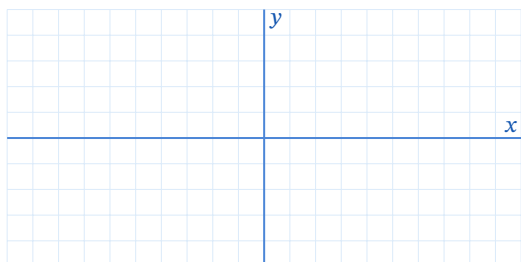
An **elementary function** is a familiar basic function whose equation, graph, domain, and range serve as useful references.

Function	Equation	Domain	Range
Constant	$f(x) = c$	$(-\infty, \infty)$	$\{c\}$
Identity	$f(x) = x$	$(-\infty, \infty)$	$(-\infty, \infty)$
Square	$f(x) = x^2$	$(-\infty, \infty)$	$[0, \infty)$
Cube	$f(x) = x^3$	$(-\infty, \infty)$	$(-\infty, \infty)$
Square root	$f(x) = \sqrt{x}$	$[0, \infty)$	$[0, \infty)$
Cube root	$f(x) = \sqrt[3]{x}$	$(-\infty, \infty)$	$(-\infty, \infty)$
Reciprocal	$f(x) = \frac{1}{x}$	$(-\infty, 0) \cup (0, \infty)$	$(-\infty, 0) \cup (0, \infty)$
Absolute value	$f(x) = x $	$(-\infty, \infty)$	$[0, \infty)$
Greatest integer	$f(x) = \lfloor x \rfloor$	$(-\infty, \infty)$	$\mathbb{Z}$

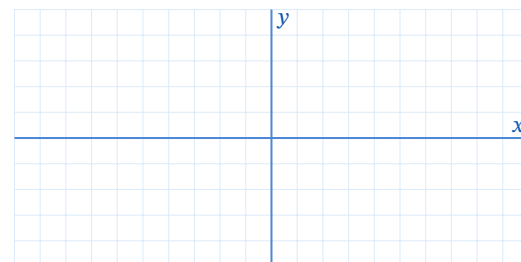
Sketch Four Elementary Functions

Use key points and the characteristic shape of each function. Keep each graph in its own coordinate plane.

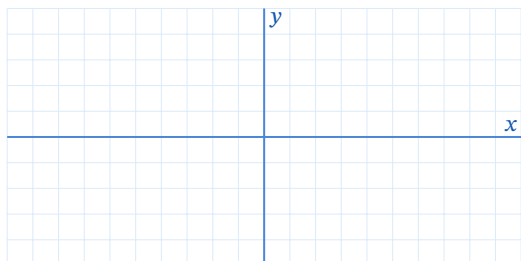
$$f(x) = x^2$$



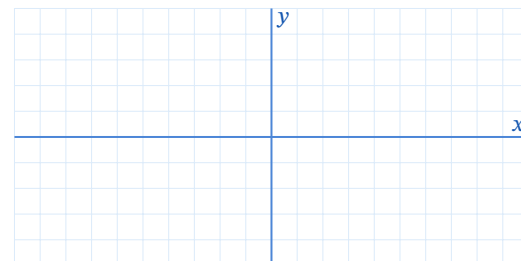
$$g(x) = x^3$$



$$h(x) = \sqrt[3]{x}$$



$$p(x) = \sqrt{x}$$



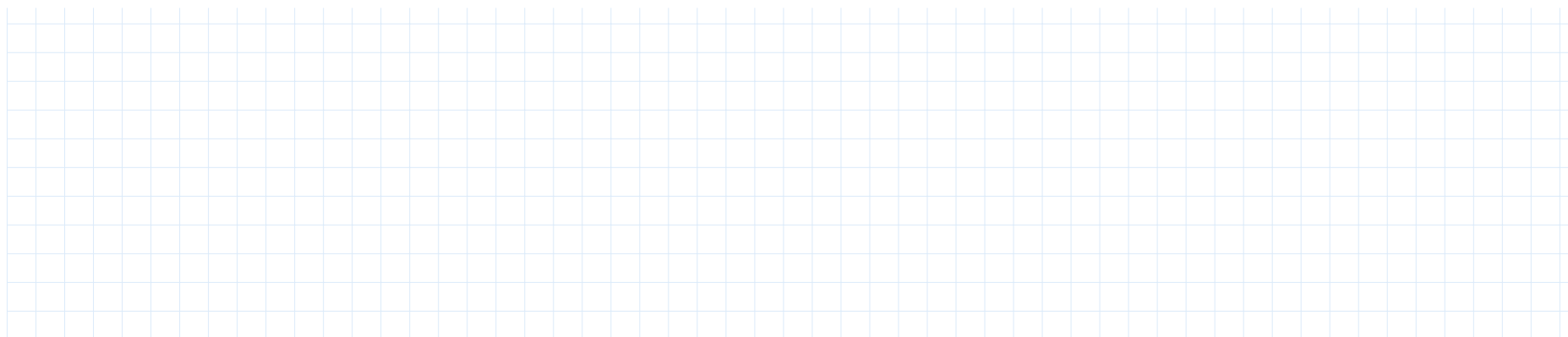
Definition

A function is **piecewise-defined** when different formulas are used on different intervals of the domain. Choose the formula whose condition contains the input. A closed endpoint is included; an open endpoint is excluded.

Let

$$f(x) = \begin{cases} x + 2, & x < 1, \\ x^2 - 1, & x \geq 1. \end{cases}$$

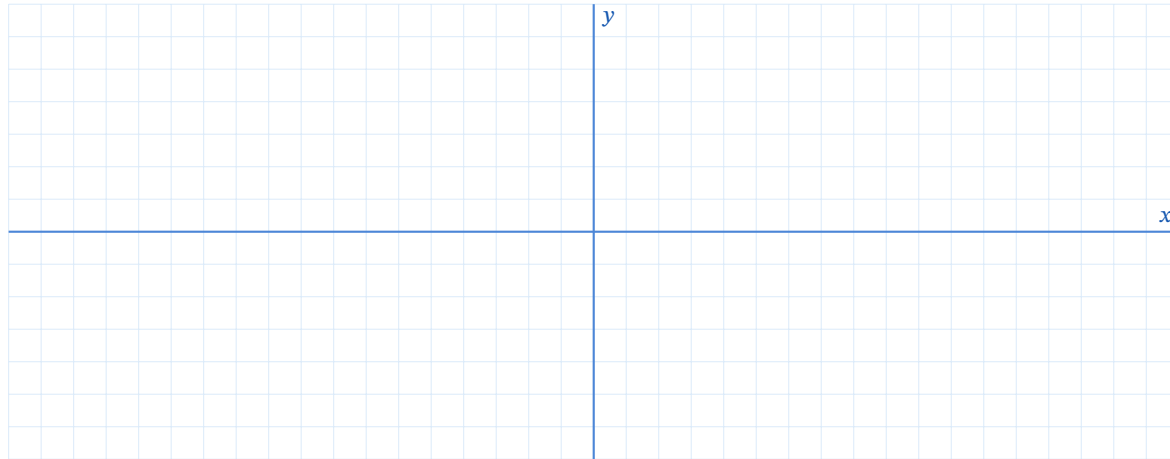
- Find $f(-3)$, $f(0)$, $f(1)$, and $f(3)$.
- For each input, explain which formula applies.
- Predict what happens at the boundary $x = 1$.



Graph a Piecewise Function

Graph both pieces and mark the boundary points correctly.

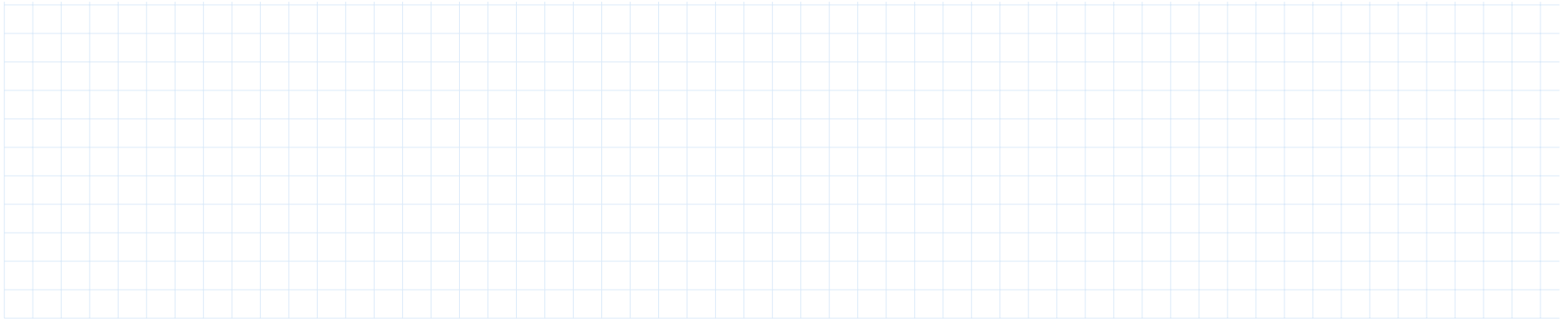
$$g(x) = \begin{cases} -2, & x < -1, \\ x + 1, & -1 \leq x < 2, \\ 4 - x, & x \geq 2. \end{cases}$$



Use the function from the previous page.

$$g(x) = \begin{cases} -2, & x < -1, \\ x + 1, & -1 \leq x < 2, \\ 4 - x, & x \geq 2. \end{cases}$$

- Evaluate $g(-4)$, $g(-1)$, $g(0)$, $g(2)$, and $g(5)$.
- Find the x - and y -intercepts.
- State the domain and range.
- State where g is increasing, decreasing, or constant.



Starting with an elementary function $y = f(x)$, a transformed graph can be written as

$$y = a f(b(x - h)) + k$$

Feature	Symbol	Effect on the graph
Vertical shift	k	$k > 0$: up; $k < 0$: down
Horizontal shift	h	$x - h$: right h units; $x + h$: left h units
Vertical scale	a	$ a > 1$: stretch; $0 < a < 1$: compression
Horizontal scale	b	$ b > 1$: compression by $\frac{1}{ b }$; $0 < b < 1$: stretch by $\frac{1}{ b }$
Reflection	sign	$a < 0$: across the x -axis; $b < 0$: across the y -axis

Reading order

First identify the elementary function. Then read horizontal changes inside $f()$ and vertical changes outside $f()$.

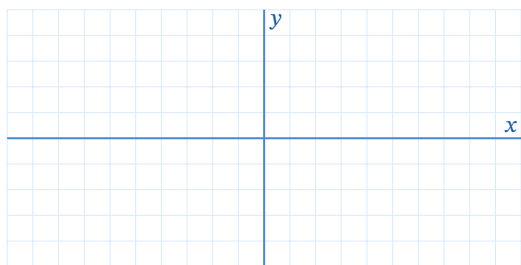
Vertical and Horizontal Shifts

Vertical shifts

$$y = f(x) + k$$

$k > 0$ moves the graph up; $k < 0$ moves it down.

$$g(x) = |x - 3| + 2$$

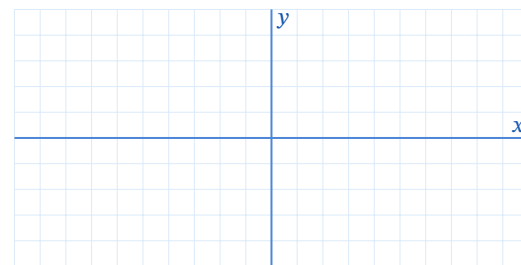


Horizontal shifts

$$y = f(x - h)$$

$h > 0$ moves the graph right; replacing h by $-h$ moves it left.

$$h(x) = \sqrt[3]{x + 2} - 1$$



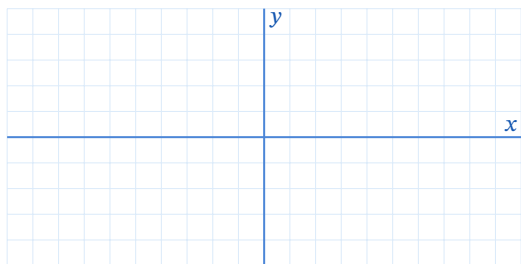
Reflections

Across the x -axis

$$y = -f(x)$$

Each output changes sign: $(x, y) \mapsto (x, -y)$.

$$g(x) = -|x|$$

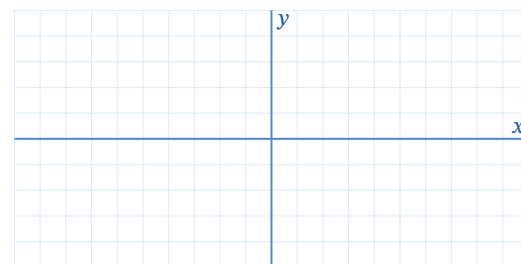


Across the y -axis

$$y = f(-x)$$

Each input changes sign: $(x, y) \mapsto (-x, y)$.

$$h(x) = \sqrt[3]{-x}$$



Vertical Stretch and Compression

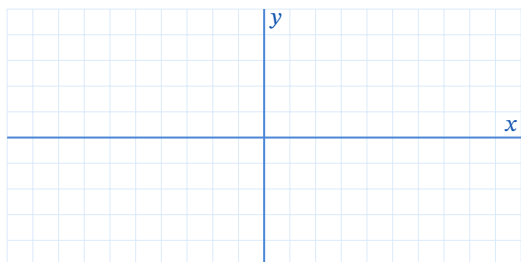
For $y = af(x)$, every y -coordinate is multiplied by a .

$|a| > 1$: vertical stretch

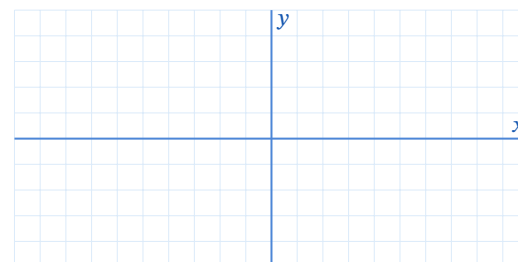
$0 < |a| < 1$: vertical compression

Use $f(x) = x^2$ and transform the key points of f before sketching.

$$y = 2f(x)$$



$$y = \frac{1}{2}f(x)$$



Horizontal Stretch and Compression

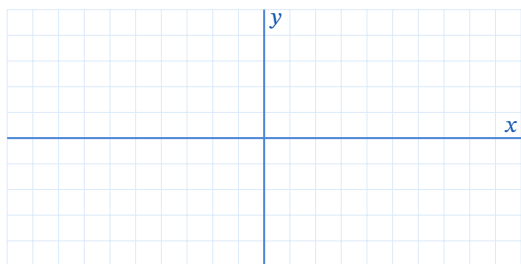
For $y = f(bx)$, each x -coordinate is divided by b .

$|b| > 1$: horizontal compression by $\frac{1}{|b|}$

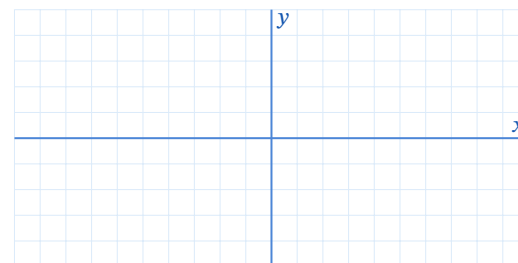
$0 < |b| < 1$: horizontal stretch by $\frac{1}{|b|}$

Again use $f(x) = x^2$ and transform the key points before sketching.

$$y = f(2x)$$



$$y = f\left(\frac{x}{2}\right)$$



Combine Transformations

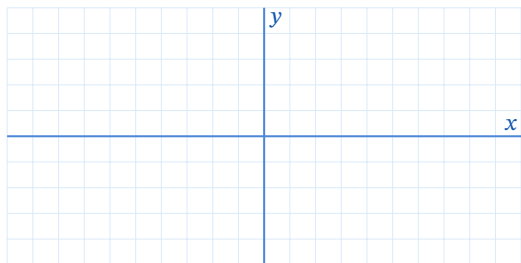
Section 1.5

Start with $f(x) = |x|$. For each function, identify the transformations, locate the new vertex, and sketch the graph.

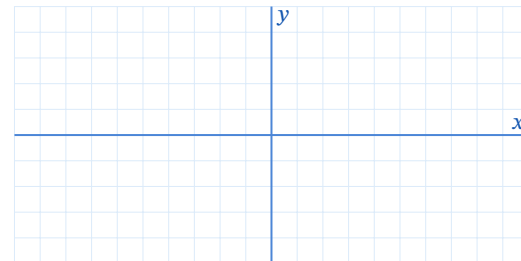
$$g(x) = -\frac{1}{2}|x - 2| + 3$$

$$h(x) = 2|x + 1| - 4$$

$$g(x) = -\frac{1}{2}|x - 2| + 3$$



$$h(x) = 2|x + 1| - 4$$



Mixed Transformation Challenge

Graph the function from its elementary function.

$$y = -2\sqrt{x+3} + 4$$

- Name the elementary function.
- List the transformations in a sensible order.
- Transform at least four useful points.
- State the domain and range.
- Identify the endpoint and any intercepts visible from the graph.

